GATE

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CE : CIVIL ENGINEERING
Module - 6 : Engineering Mechanics

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Engineering Mechanics

INTRODUCTION AND BASIC CONCEPTS

Definition of Mechanics :
The science which deals with state of rest or motion of a body, forces acting on the body and the effects of these forces on the bodies upon which they act.

Engineering Mechanics :
The study of the laws of Mechanics as applied to engineering. It is mainly concerned with the application of the principles of Mechanics in solving engineering problems.

Different Branches of the Subject :
1) Statics : Deals with the forces acting upon rigid bodies and their effects, when the body is at rest.
2) Dynamics : It is that part of the subject which deals with particles and bodies that are in motion. It is subdivided into kinematics and kinetics.
   i) Kinematics : It is the study of the motion of a body without consideration of the mass of the body or the forces causing motion in the body.
   ii) Kinetics : It is the study of the motion of a body with all its changes, due to inertia of masses acted upon by the forces.

Force and Force Systems :
Force may be defined as an action (push or pull) that changes or tends to change the state of rest or of uniform motion of the body upon which it acts.

Systems of Forces :
If a number of forces act simultaneously on a body, they are said to form, what is called, a system of forces. Depending upon whether all the forces acting on the body lie in the same plane or in different planes, the systems of forces may be classified as :
i) Co–planar system in which all the forces lie in the same plane
ii) Non co–planar system in which all the forces do not lie in the same plane but lie in different planes.

These two types can be further subdivided into :
a) Collinear Forces : Forces which act along the same straight line
b) Concurrent Forces : Forces whose lines of action meet at a point
c) Parallel Forces : Force acting along parallel lines
d) Non Concurrent Non Parallel Forces : Forces acting along lines which are neither parallel nor do these meet at a point.

Resultant and Components :
If any number of forces acting upon a body can be replaced by a single force which shall have the same effect as all the forces together then this single force is called as resultant force and the forces of which this is the resultant are called the components of their resultants.

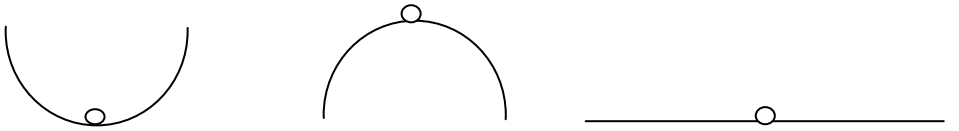
Kinds of Equilibrium :
Equilibrium of a body means a state of rest or balance under the effect of the external forces acting on the body. Depending upon the position of the resultant force acting on a body, its equilibrium may be of any one type of the following three kinds of equilibrium.

i) Stable Equilibrium :
A body is said to be in a stable equilibrium if, given a slight displacement it tends to return to its original position.

ii) Unstable Equilibrium :
A body is said to be in unstable equilibrium if, given a slight displacement it tends to keep moving along the direction of the displacement and never returns to its original position.

iii) Neutral Equilibrium :
A body is said to be in neutral equilibrium if on being given a small displacement, it neither returns to its original position nor loses its equilibrium. The body now attains a new equilibrium position.

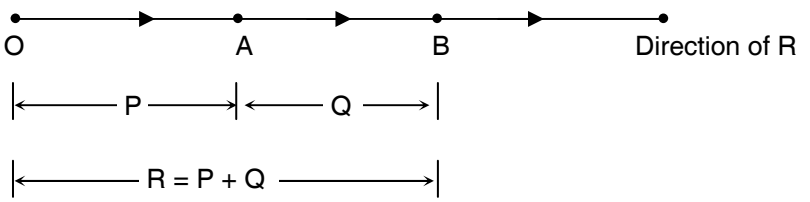
The figure below shows the three different kinds of equilibrium.



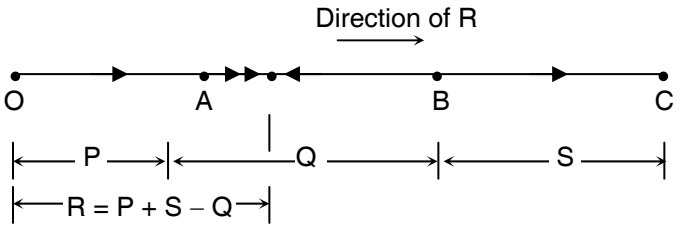
a) Stable equilibrium b) Unstable equilibrium c) Neutral equilibrium

COMPOSITION AND RESOLUTION OF FORCES

Resultant of Forces Acting in a Straight Line :
The resultant of any number of forces acting in the same direction is equal to their sum and acts in the same direction as the forces as shown.

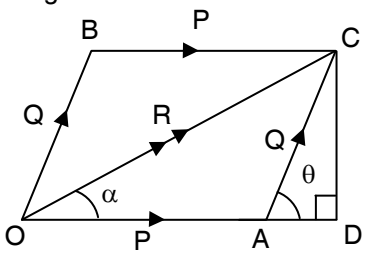


If all the forces do not act in the same direction, some acting in one direction, the others acting in the opposite direction, then the resultant is equal to the difference between the resultant of those acting in one direction and the resultant of those which act in the opposite direction and has the direction of the greater of the two resultant as shown :



Resultant of two Concurrent Forces :

Let us now consider figure below :



Here the adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent the forces $\vec{P}$ and $\vec{Q}$ respectively.

AC = OB = Q

CD = Q sin θ, AD = Q cos θ

OD = OA + AD

= P + Q cos θ

$R^2 = CD^2 + OD^2$

$CD^2 = Q^2 \sin^2\theta$, $OD^2 = P^2 + Q^2\cos^2\theta + 2PQ\cos\theta$

$\therefore R^2 = P^2 + Q^2 + 2PQ\cos\theta$

Thus the magnitude of the resultant $\vec{R}$ is given by

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

where θ is the angle made by $\vec{Q}$ with $\vec{P}$.

The direction of resultant is given by

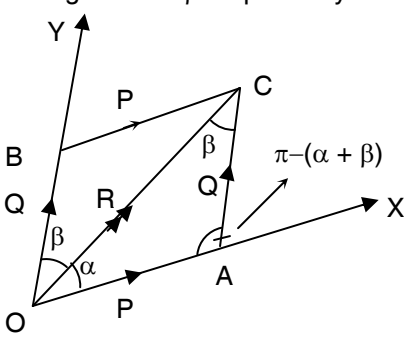
$$\alpha = \tan^{-1}\left(\frac{Q\sin\theta}{P + Q\cos\theta}\right)$$

where α is the angle made by $\vec{R}$ with $\vec{P}$.

Components of a Given Force in Two Given Directions :

Let $\overrightarrow{OC}$ represents a force which is required to be resolved into two components P and Q acting along known directions OX and OY respectively.

Let OX and OY make angle α and β respectively with $\overrightarrow{OC}$ as shown in the figure.



Applying sine formula to Δ OAC,

$$\frac{OA}{\sin\beta} = \frac{AC}{\sin\alpha} = \frac{OC}{\sin\left[\pi - (\alpha + \beta)\right]}$$

$$\Rightarrow \frac{P}{\sin\beta} = \frac{Q}{\sin\alpha} = \frac{R}{\sin(\alpha+\beta)}$$

i.e $$P = \frac{R\sin\beta}{\sin(\alpha+\beta)}$$

$$Q = \frac{R\sin\alpha}{\sin(\alpha+\beta)}$$

A special case :

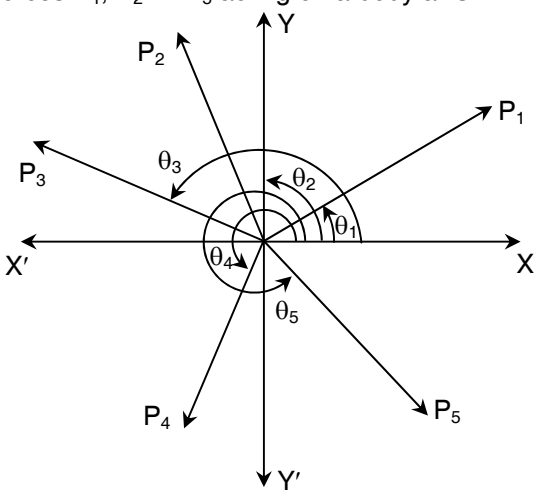
Let $\alpha + \beta = 90°$, then

$P = R\cos\alpha$ (or $R\sin\beta$)

$Q = R\sin\alpha$ (or $R\cos\beta$)

Resultant of a Number of Co–planer Concurrent Forces :

Consider the forces P_1, P_2 … P_5 acting on a body at O.



Let XOX′ and YOY′ be any two reference axes, taken perpendicular to each other and passing through O. let these forces make angles $\theta_1, \theta_2, \theta_3, ...\theta_5$ respectively with XOX′ as shown in the figure. Let R be the resultant force inclined at an angle θ to XOX′. Then, by principle of resolved parts :

$R\cos\theta = P_1\cos\theta_1 + P_2\cos\theta_2 + . . . + P_5\cos\theta_5$ (for equilibrium in horizontal direction)

$= \Sigma H$ (say X) …(1)

and $R\sin\theta = P_1\sin\theta_1 + P_2\sin\theta_2 + ... + P_5\sin\theta_5$ (fir equilibrium in vertical direction)

$= \Sigma V$ (say Y) …(2)

By squaring and adding the above two equations we get,

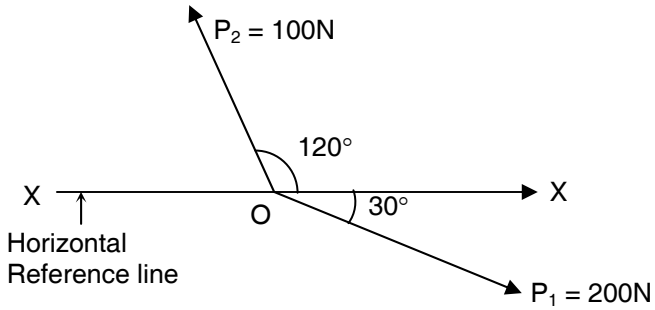
$$R = \sqrt{X^2 + Y^2}$$

Also dividing equation (2) by (1) we get ;

$$\frac{R\sin\theta}{R\cos\theta} = \tan\theta = \frac{Y}{X} \Rightarrow \theta = \tan^{-1}\left(\frac{Y}{X}\right) = \tan^{-1}\left(\frac{\Sigma V}{\Sigma H}\right)$$

Example :

Two forces P_1 and P_2 act on a particle as shown. Calculate the resultant force on the particle



Solution :

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

Here $P = 200$ N ($= P_1$)

$Q = 100$ N ($= P_2$)

$\theta = 120° + 30° = 150°$

$$R = \sqrt{200 + 100 + 2 \times 200 \times 100 \cos 150°} = 124\text{N}$$

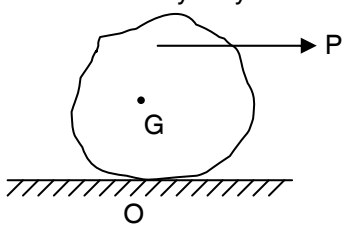
Let the resultant makes angle α with P_1

then $$\alpha = \tan^{-1}\left(\frac{Q\sin\theta}{P + Q\cos\theta}\right) = \tan^{-1}\left(\frac{100 \times \sin 150}{200 + 100 \times \cos 150}\right)$$

$$= 23°46'$$

Moments :

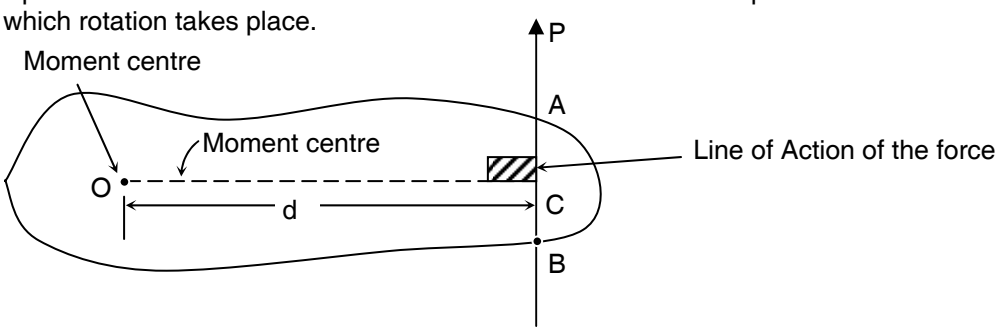
Consider a force P acting on a body at rest as shown. As the force does not pass through G, the centre of gravity of the body, it will not given the body a straight line motion, called the translatory motion, but will tend to rotate it about O. The measure of this tendency of a force by virtue of which it tends to rotate the body on which it acts is called the moment of the force. This rotation of the body may be either about a point or a line.



The moment of a force about a point or a line depends upon :

i) The magnitude of the force and

ii) The perpendicular distance of the line of action of the force from the point or the line about which rotation takes place.



Thus referring to the figure if P(in Newtons) is the force acting on the body along AB and d, the perpendicular distance of AB from O (in metres) called the moment centre or fulcrum, then

$$K = \text{moment of the force AB along O} = P \times d = P \times OC$$

The distance p is termed as moment arm.

$\therefore$ Moment of a force = force $\times$ perpendicular distance between the line of action of the force and the point or the axis about which moment is required.

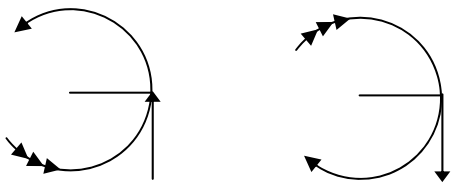
Different Types of Moments :

The moments of a force may be classified depending upon :

i) The direction of rotation caused by a force, and

ii) The effect produced due to rotation caused in the body.

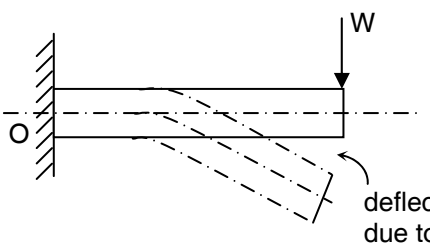
Depending upon the direction of rotation, the moment may be a clockwise moment or an anti–clockwise moment. The moments is said to be clockwise if the tendency of the force is to cause rotation in the body in the same direction in which hands of the clock move and anti–clockwise if the force tends to rotate the body in a direction opposite to that of the hands of a clock. Conventionally, an anti–clockwise moment is always taken as positive and clock moment as negative. To remember this conversion, one can curl the fingers of the right hand along the direction of the moment. If, upon doing this the thumb finger points outside the plane of the paper, the moment is positive and if it points into the plane of the paper, the moment is negative. Figure below makes this distinction clear.



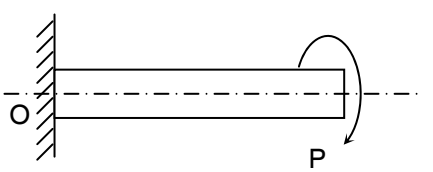
Anticlockwise Moment

Clockwise Moment

To classify the moments depending upon the effect produced, consider a weight W acting on a cantilever at its free end as shown in the figure. The effect of this load is to cause bending deflection of the cantilever about O and hence the moment due to W is called bending moment. However, if a force P causes to twist the shaft as shown in the figure then the moment due to P is called the twisting moment.



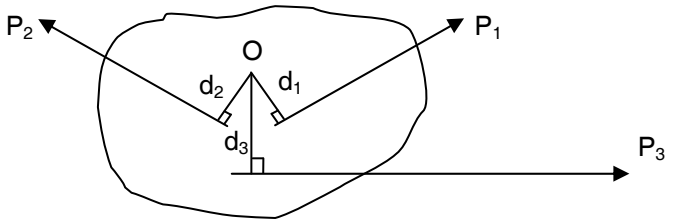
Bending Moment



Twisting Moment

Resultant Turning Moment :

Consider the forces P_1, P_2, P_3 acting on a body simultaneously as shown in the figure. Let d_1, d_2, d_3 be their respective distances from a point O in the body. The force P_1, has an anticlockwise moment P_1d_1 about O, the force P_2 has the moment P_2d_2 (clockwise) whereas the force P_3 has the moment $P_3\,d_3$(anti–clockwise).



$\therefore$ Net moment acting on the body

= Difference of anti–clockwise and clockwise moments

$= \left(P_1d_1 + P_3d_3\right) - P_2d_2$

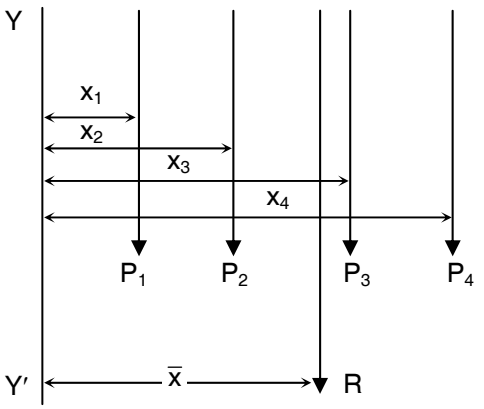
Principle of Moments – Varignon's Principle :

It states that the algebric sum of the moments of a given system of forces about a point is equal to the moment of their resultant about the same point.

Resultant of Co–Planar Parallel Forces :

When a number of parallel forces act in a plane, their resultant may either be a force or a couple. In the figure shown, all the force P_1, P_2, … P_4 are parallel and their lines of action are at distances x_1, x_2, x_3, x_4 respectively from a reference line YY′. The magnitude of the resultant R of the forces P_1, … P_4 will be the algebraic sum of the forces.

$R = P_1 + P_2 + P_3 + P_4$



Now the point of action of the resultant may be obtained by taking moments about the reference line YY′. If we let $\bar{x}$ be the distance of R from YY′, then by using the principle of moments,

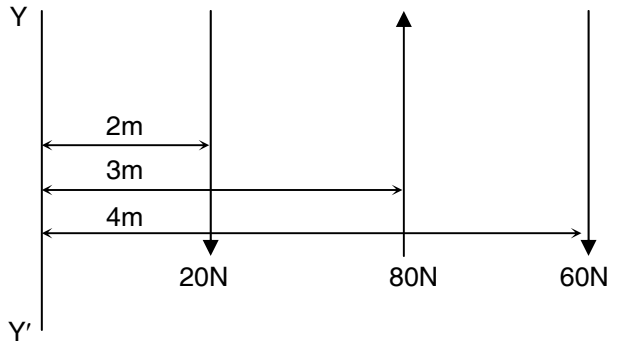
$$R\bar{x} = P_1x_1 + P_2x_2 + P_3x_3 + P_4x_4$$

$$\Rightarrow \quad \bar{x} = \frac{P_1x_1 + P_2x_2 + P_3x_3 + P_4x_4}{P_1 + P_2 + P_3 + P_4} \qquad \because R = P_1 + P_2 + P_3 + P_4$$

$$= \frac{\sum P_ix_i}{\sum P_i}$$

Example :

Show that the force system, shown in the figure, has a zero resultant but still the system reduces to a couple. Also find the moment of this couple.



Solution :

$R = P_1 + P_2 + P_3 = 80 - 20 - 60 = 0$ N

(Taking upward direction positive)

Taking moments of all the forces about YY′ and taking anti–clockwise moments as positive :

Resulting moment $M = -20 \times 2 + 80 \times 3 - 60 \times 4$

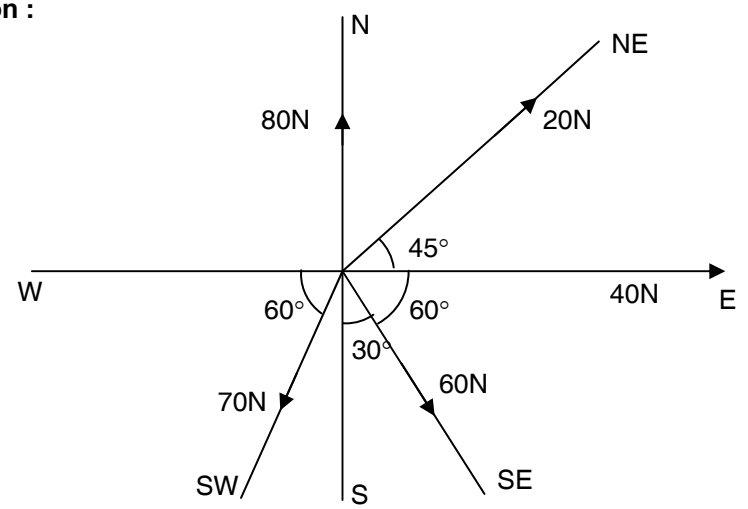
$= -40$ Nm

Example :

Find the magnitude of the resultant of the following forces acting at a point.

i) 80 N due north
ii) 20 N due north east
iii) 40 N due east
iv) 60 N in a direction inclined 30° east of south
v) 70 N in a direction inclined 60° south of west.

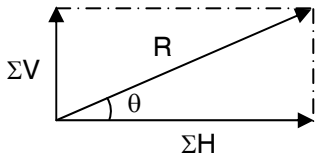
Solution :



$\Sigma H = 0 + 20 \cos 45° + 40 + 60 \cos 60° + (-70 \cos 60°)$

$= 49.142$ N (East wards)

$\Sigma V = 80 + 20 \sin 45° + 0 - 60 \sin 60° - 70 \sin 60°$
$= -18.438$ N (Northwards)



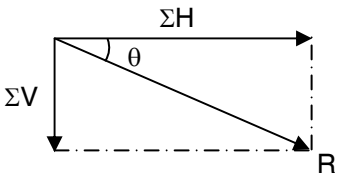
$R = \sqrt{(\Sigma V)^2 + (\Sigma H)^2}$ (Since ΣV is –ve so actually figure will be)

$= \sqrt{(49.142)^2 + (-18.438)^2}$

$= 52.487$ N

$\tan\theta = \dfrac{\Sigma V}{\Sigma H} = 0.375$

$\theta = 20.5°$

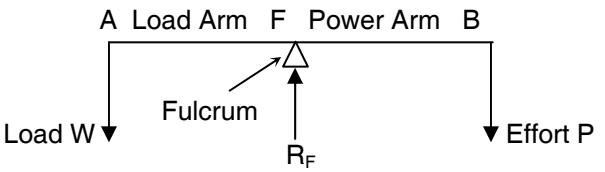


Law of Moments :
It states that if a number of coplanar forces, act upon a body and produce equilibrium, then the sum of the clockwise moments is equal to the sum of the anti–clockwise moments or the algebraic sum of moments taken about any point in the plane of the forces is zero.

Levers :
A lever is a rigid bar, straight, bent or curved which is free to or able to rotate about a fixed point in it, called the fulcrum of the lever.

Mechanical Advantage of a Lever :
Consider the lever shown in the figure in which P is the effort applied to raise the load W, R_F being the reaction at the fulcrum F. From the principle of moments taking moments about point F,

$P \times FB = W \times FA$



or $\dfrac{W}{P} = \dfrac{FB}{FA}$ and is called the mechanical advantage or the leverage of the lever (M. A.).

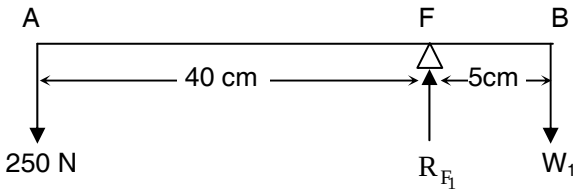
Thus M. A. of a lever = $\dfrac{\text{Power Arm}}{\text{Load Arm}}$

Example :

A light horizontal lever AB of length 45 cm is pivoted about the fulcrum F, 5cm from B. An effort of 250 N is applied vertically at the end A. Find the reaction at F and the resistance (W_1) overcome at B. If W_1 and F change place, find the reaction at F and the resistance (W_1) overcome.

Solution :

(i) Taking moments about F :



$$250 \times 40 = W_1 \times 5$$

$$\Rightarrow \quad W_1 = \frac{250 \times 40}{5} = 2000N$$

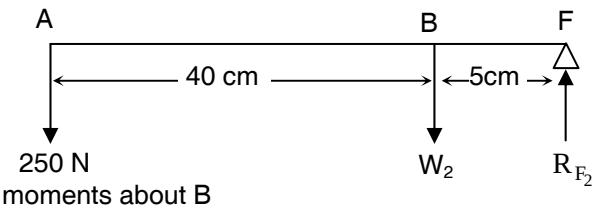
For equilibrium in vertical direction,

$250 + W_1 = R_{F_1}$,

where R_{F_1} reaction at F. Thus,

R_{F_1} = 2250 N.

(ii) When W and F change place, the forces acting on the lever are as shown :



Taking moments about B

$$250 \times 40 = R_{F_2} \times 5$$

$$\Rightarrow \quad R_{F_2} = \frac{250 \times 40}{5} = 2000N$$

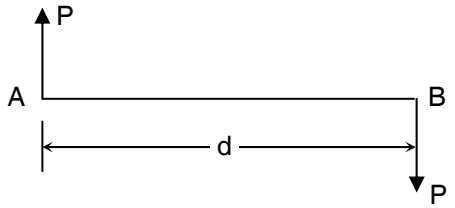
Also, taking moments about F

$$W_2 \times 5 = 250 \times 45$$

$$\Rightarrow \quad W_2 = \frac{250 \times 45}{5} = 2250N$$

Couple :

A couple consists of two equal and parallel forces acting in opposite directions so that their resultant is zero and hence a couple has no tendency to produce translatory motion but produces rotation in the body on which it acts. Figure below shows two equal and opposite forces each having a magnitude equal to P and acting at A and B along parallel lines, thus constituting a couple. The perpendicular distance AB is called the arm of the couple and is denoted by d. The couple shown in the figure is conventionally written as a couple $P \times d$.



Moment of a Couple :

The tendency of a couple to cause rotation in the body on which it acts is measured by the product of either of the forces constituting the couple and the arm of the couple. This product $P \times d$ is called moment of the couple.

Two Types of Couples :

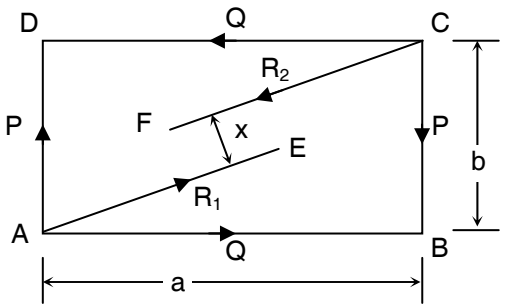
When the tendency of the couple is to rotate the body in the anti–clockwise direction it is called an anti-clockwise or a positive couple whereas a couple having tendency to rotate the body in the clockwise direction is taken as a clockwise or a negative couple.

Example :

ABCD is a rectangle such that AB = CD = a and BC = DA = b. Forces of magnitude equal to P act along AD and CB and magnitude equal to Q act along AB and CD. Prove that the perpendicular distance between the resultant of P, Q at A and that of P, Q at C is

$$\frac{Qb - Pa}{\sqrt{P^2 + Q^2}}$$

Solution :



Here, $R_1 = R_2 = \sqrt{P^2 + Q^2}$

Moment of couple due to forces P = P × a (clockwise) and moment of the couple due to the forces Q = Q × b (anti clockwise)

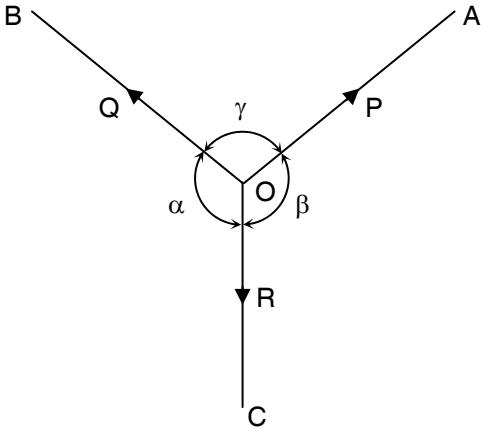
∴ Resultant moment = Qb – Pa (anti clockwise) and must be equal to the moment of the resultant $R_1 = R_2 = R$. If x is the perpendicular distance between the lines of actions of R_1 and R_2 then

$$R \times x = Qb - Pa$$

$$\Rightarrow \quad x = \frac{Qb - Pa}{R} = \frac{Qb - Pa}{\sqrt{P^2 + Q^2}}$$

Lami's Theorem :

It states that if three forces acting at a point are in equilibrium, then each is proportional to the sine of the angle between the other two. Let P, Q and R be three forces represented by OA, OB and OC as shown in the figure.



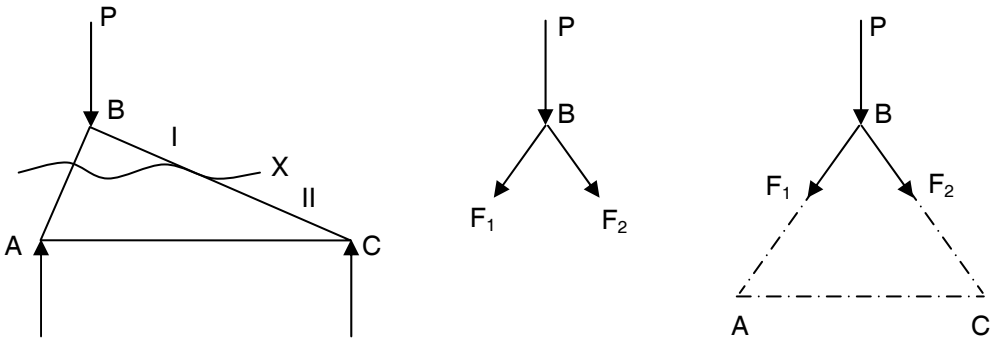
Thus according to Lami's theorem,

$$\frac{P}{\sin\alpha} = \frac{Q}{\sin\beta} = \frac{R}{\sin\gamma}$$

ANALYSIS OF TRUSSES AND FRAMES

Free Body Diagrams :

To solve a problem of finding the forces in a structure, each member is separated from the rest of the structure and represented in a separate sketch. The forces exerted on this member by the other members are shown in this sketch.



(a) (b) (c)

Such a sketch is called a free body diagram. In case of a beam or truss mounted on supports, the reactions which the supports exert on the beam, are included as external forces in the free body diagram. In the above figure (a), the supports at A and C are not shown explicitly but are replaced by equivalent reactions. To find the stresses in the members of a truss, each member is isolated and considered as a free body diagram. The effect of the part removed are introduced as external forces acting on the free body under analysis and the free body diagram is assumed to be in equilibrium under the forces acting on it alone.

In figure (a), the truss is divided by the section into parts I and II and the portion on the bottom of the section is removed leaving behind that shown in the figure (b). This is the

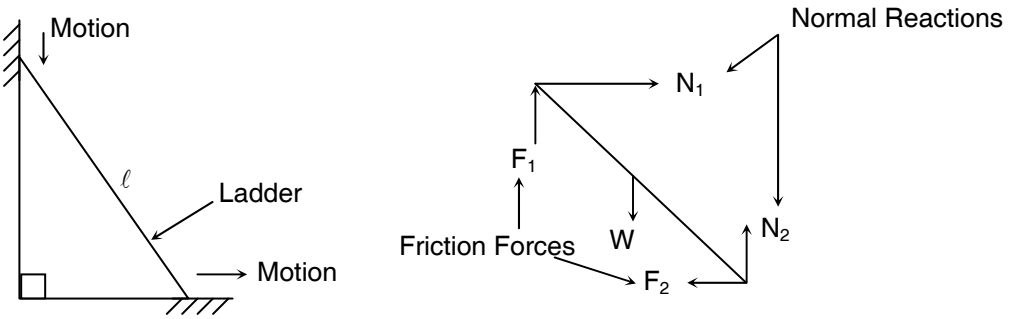
free body diagram as the effects of the truss shown in the figure (b). In marking the directions of these external forces on the cut members BA and BC, we shall always start with the assumption that these are tensile forces i.e. act away at the joint (B here) along the cut members (BA and BC here) and if after calculating these come to be negative, it will mean that our assumption of a tensile force is not correct and the particular member having a negative value of the truss is under compression and hence the direction of the force on this member should be reversed.

Figure below shows a member under tension and a member under compression:

Free body concept is also applied to other types of problems :

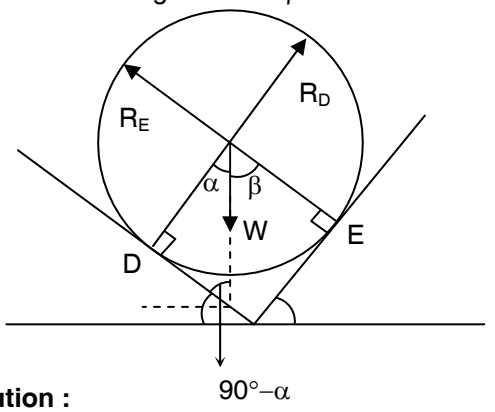
1. Figure a below shows a solid sphere of radius R and weight W, resting against rectangular block. A force P acts in the horizontal direction on sphere. For the above situation, the free body diagram of the sphere is shown in fig. b, where N is reaction force.

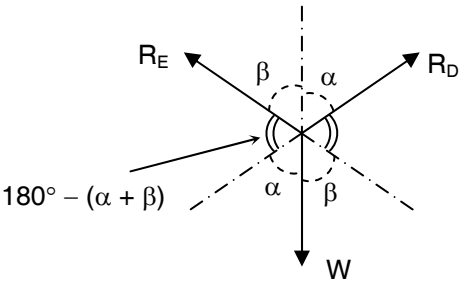
2. Consider a ladder of length ℓ and weight W is kept at an inclined position with respect to a vertical wall. As the ladder begins to slide down, there are frictional forces F_1, F_2 and reaction forces N_1, N_2 acting on the ladder. The figure below shows the situation and the free body diagram of the ladder.



Example 1 :

A heavy Spherical ball of weight W rests in a V shaped trough whose sides are inclined at angles α and β to the horizontal. Find the reaction on each side.





Solution :

Applying Lami's Theorem :

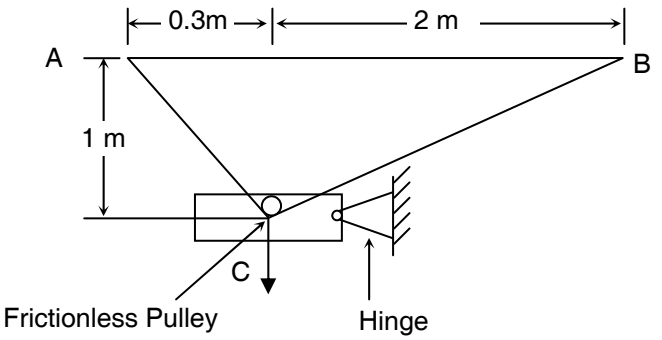
$$\frac{R_E}{\sin\left[\beta+180°-(\alpha+\beta)\right]}=\frac{R_D}{\sin\left[\alpha+180°-(\alpha+\beta)\right]}=\frac{W}{\sin(\alpha+\beta)}$$

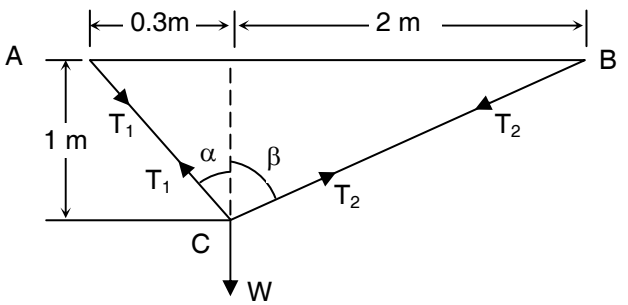
$$\Rightarrow \quad \frac{R_E}{\sin(180°-\alpha)}=\frac{R_D}{\sin(180°-\beta)}=\frac{W}{\sin(\alpha+\beta)}$$

$$\Rightarrow \quad R_E=\frac{W\sin\alpha}{\sin(\alpha+\beta)} \text{ and } R_D=\frac{W\sin\beta}{\sin(\alpha+\beta)}$$

Example 2 :

Find the forces in the cables AC, BC for the system of suspension as shown.





$$\alpha = \tan^{-1}\left(\frac{0.3}{1}\right) = 16°42'$$

$$\beta = \tan^{-1}\left(\frac{2}{1}\right) = 63°26'$$

Applying Lami's Theorem :

$$\underset{\text{(angle between } T_2 \text{ and W)}}{\frac{T_1}{\sin\left[\alpha + 180° - (\alpha+\beta)\right]}} = \underset{\text{(angle between } T_1 \text{ and W)}}{\frac{T_2}{\sin\left[\beta + 180° - (\alpha+\beta)\right]}}$$

$$= \frac{W}{\sin(\alpha+\beta)}$$

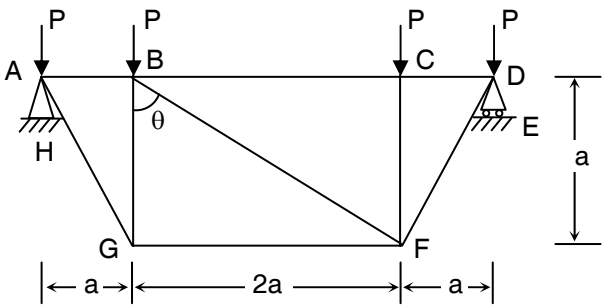
Substituting values of W, α and β we get ;

T_1 = 0.908 WN

T_2 = 0.292 WN

Example 3 :

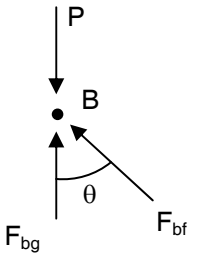
Find the force in member BF of truss as shown



Solution :

Analysing Joint B;

FBD of Joint B is shown below :



Now summing all vertical forces :

$P - F_{bg} - F_{bf} \cos\theta = 0$ …(1)

But, summing all horizontal forces,

$F_{bf} \sin\theta = 0$ …(2)

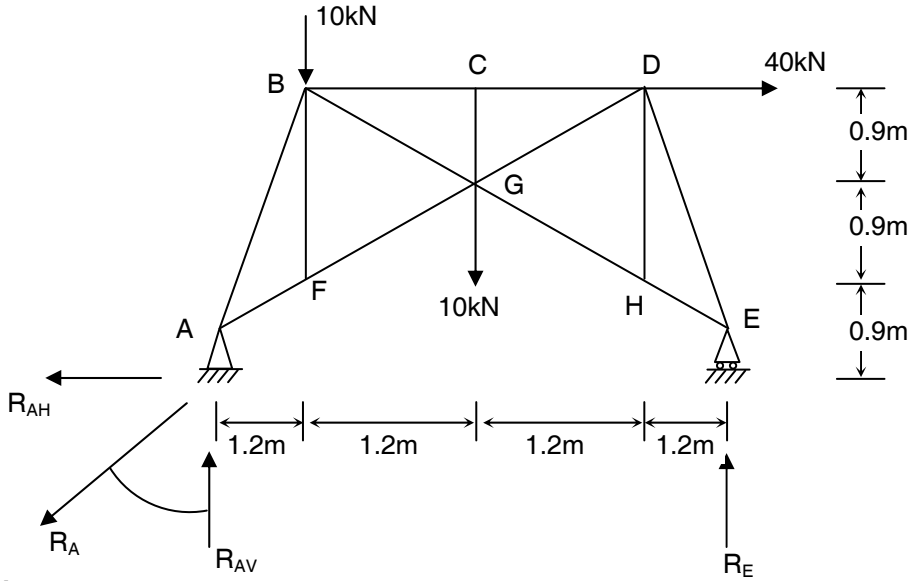
Hence from (2); $F_{bf} = 0$ $(\sin\theta \neq 0)$

and hence $P = F_{bg}$

Force in member BF is zero.

Example 4 :

The pin connected truss ABCDE is shown in the figure. Determine the magnitude and direction of the reactions at the two supports, A and E.



Solution :

Taking moments about A;

$$R_E \times (1.2 + 1.2 + 1.2 + 1.2) = 10 \times 1.2 + 10 \times (1.2 + 1.2) + 40 \times (0.9 + 0.9 + 0.9)$$

$\Rightarrow$ R_E = 30 kN

Now Σ H = 0 gives : R_{AH} = 40 kN

Σ V = 0 gives :

R_{AV} + 10 + 10 – 30 = 0

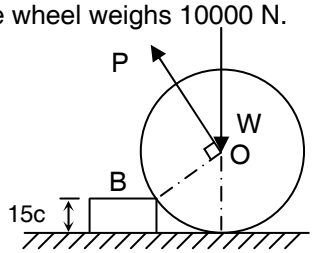
$\Rightarrow$ R_{AV} = 10 kN

Hence $R_A = \sqrt{R_{AV}^2 + R_{AH}^2} = \sqrt{10^2 + 40^2} = 41.23\text{kN}$

Also $\theta = \tan^{-1}\left(\frac{R_{AH}}{R_{AV}}\right) = \tan^{-1}\left(\frac{40}{10}\right) = \tan^{-1}(0.4) = 76°$.

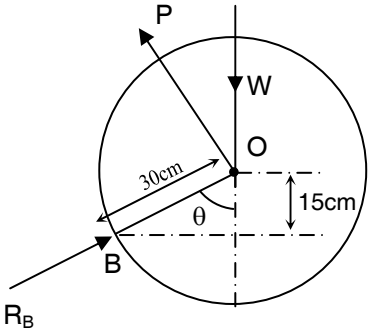
Example 5 :

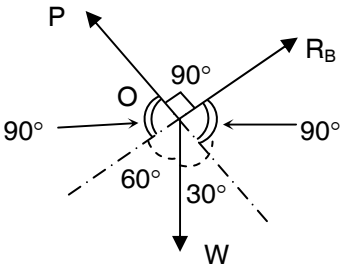
A uniform wheel 60 cm in diameter rests against a rigid rectangular block 15 cm thick as shown in the figure. Find the least pull P through the centre of the wheel to just turn the wheel over the corner of the block. All surfaces are smooth. Find also reaction of the block. The wheel weighs 10000 N.



Solution :

FBD of the wheel is as shown :





Equilibrium at point 'O'

$\cos\theta = \frac{15}{30} = \frac{1}{2}$ $\Rightarrow \theta = 60°$

Applying Lami's Theorem

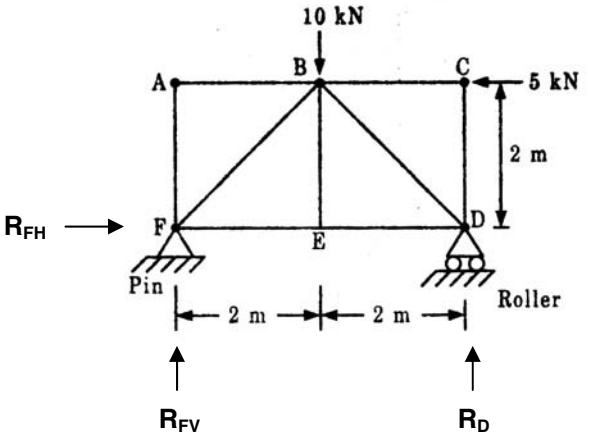
$$\frac{P}{\sin(90+30)} = \frac{W}{\sin 90} = \frac{R_B}{\sin(60+90)}$$

$\Rightarrow$ $P = W \sin 120° = \frac{W\sqrt{3}}{2} = 10000 \times \frac{\sqrt{3}}{2} = 8660N$

and $R_B = W \sin 150° = W \times \frac{1}{2} = 10000 \times \frac{1}{2} = 5000N$

Example 6 :

Find the forces in each member of the truss as shown :



Solution :

Taking moments about D ;

$10 \times 2 + 5 \times 2 = R_{FV} \times 4$

$\Rightarrow$ $R_{FV} = 7.5kN$

Applying $\Sigma V = 0$ we get

$10 = R_{FV} + R_D$

$\Rightarrow \quad R_D = 10 - R_{FV}$

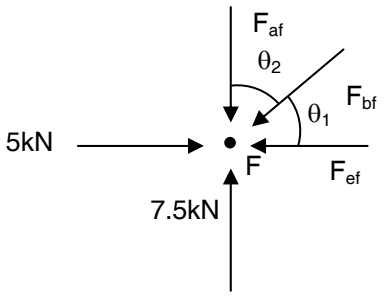
$= 10 - 7.5 = 2.5$ kN

Applying $\Sigma H = 0$ we get

$R_{FH} = 5$ kN

Now we will analyze the truss, joint by joint :

Joint F : FBD is shown below



$\theta_1 = \theta_2 = 45°$

$\Sigma H = 0$ gives

$F_{bf} \sin 45° + F_{ef} = 5\text{kN}$ …(1)

$\Sigma V = 0$ gives $7.5 \text{ kN} = F_{af} + F_{bf} \cos 45°$ …(2)

But by seeing joint A we can simply conclude that

$F_{AB} = 0$ and $F_{AF} = 0$ (no external force)

Similarly for joint E; $F_{EB} = 0$

Now; as $F_{af} = 0$; from (2)

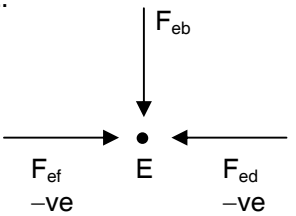
$$F_{bf} = \frac{7.5}{\cos 45°} = 10.7\text{kN} \text{ (same as assumed direction)}$$

Now from equation (1)

$F_{ef} = 5 - F_{bf} \sin 45$

$= -2.5$ kN (opposite to assumed direction)

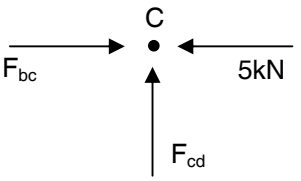
Now For joint E:



$\Sigma V = 0$ gives $F_{eb} = 0$

$\Sigma H = 0$ gives $F_{ed} = F_{ef} = -2.5$ kN

Joint C :

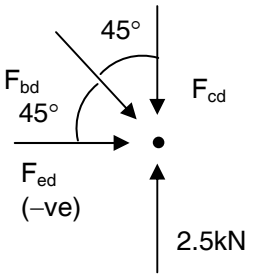


$\Sigma V = 0$ gives $F_{cd} = 0$

$\Sigma H = 0$ gives

$F_{bc} = 5$ kN (same as assumed direction)

Joint D:



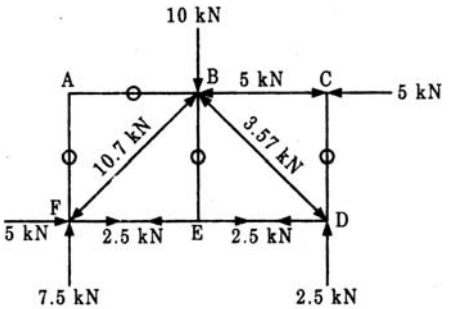
$\Sigma V = 0$ gives

$F_{bd} \cos 45° + F_{cd} = 2.5$ kN

$$\Rightarrow \quad F_{bd} = \frac{2.5}{\cos 45} \qquad (F_{cd} = 0)$$

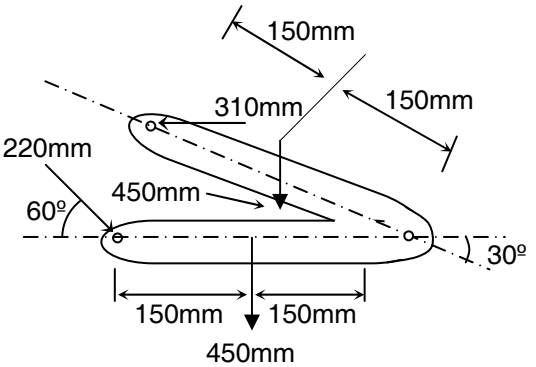
$= 3.57$ kN (same as assumed direction)

All internal forces are shown below:



Example 7 :

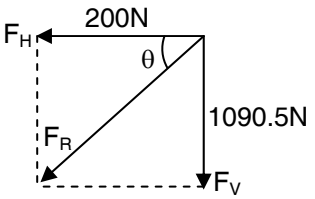
Figure below shows a bell crank lever. Find the resultant force on it.



Solution :

$\Sigma V = 450 + 450 + 220 \sin 60° = 1090.5$ N $= F_V$

$\Sigma H = 310 - 220 \cos 60° = 200$ N $= F_H$

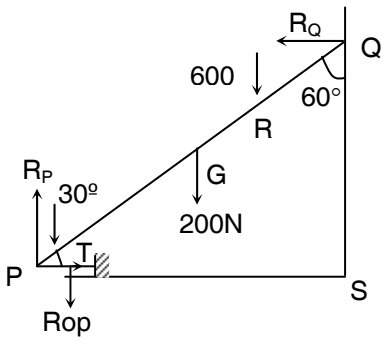


$$F_R = \sqrt{F_H^2 + F_V^2} = \sqrt{(1090.5)^2 + (200)^2} = 1108.7N$$

$$\theta = \tan^{-1}\frac{F_V}{F_H} = 79.61°$$

Example 8 :

As shown in the figure below, a man weighting 600N climbs over a ladder 5m long weighing 200N which rests at angle of 60° to the vertical with its ends resting on a smooth floor and against a smooth vertical wall, the lower end being attached by a rope as shown in the figure. The load carrying capacity of the rope is 1000N. What is the safe distance upto which the man can climb over the ladder ?



Solution :

Let us assumed that the rope fails when the man is at R on the ladder. Let PR = x meters. Tension in the rope at the time of failure = T = 1000N.

Now $\Sigma H = 0$ gives $T = R_Q = 1000N$

Taking moments about P;

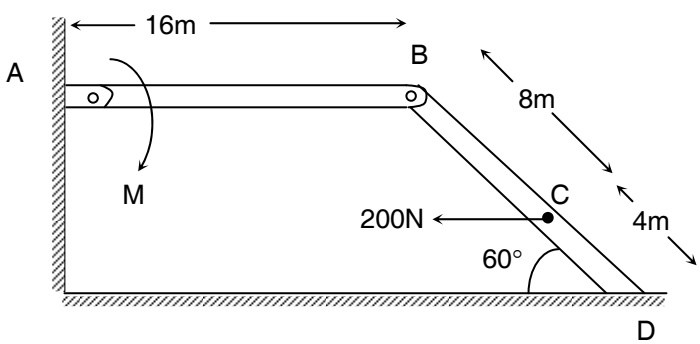
$200 \times PG \cos 30° + 600 \times PR \cos 30 = RQ \times PQ \sin 30$

$\Rightarrow \quad 200 \times 2.5 \times 0.866 + 600 \times x \times 0.86 = 1000 \times 5 \times 0.5$

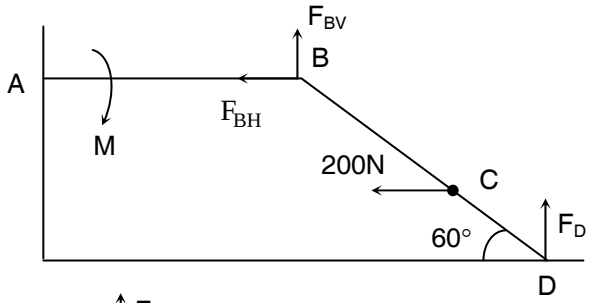
$\Rightarrow \quad x = 3.97$ m

Example 9 :

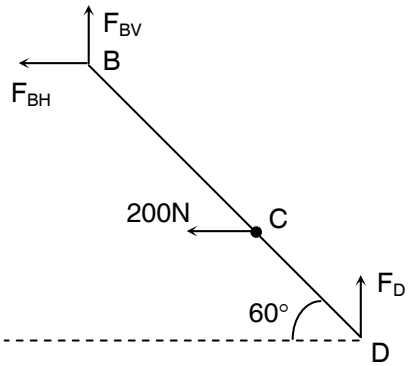
Figure shows a member BCD whose upper end is pinned at B to the horizontal member AB while bottom rests on a smooth horizontal plane. If a horizontal force of 200N is applied to the member BCD, then what moment M must be applied to the member AB to hold the system in equilibrium? What is the magnitude of the pin reaction of 'B' ?



Solution :



FBD of BCD :



$\Sigma V = 0$

$\Rightarrow \quad F_{BV} + F_D = 0$ …(1)

$\Sigma H = 0$

$\Rightarrow \quad 200 + F_{BH} = 0, F_{BH} = -200\text{N}$ …(2)

Taking moments about D; $\Sigma M_D = 0$;

$F_{BV} \times 12\cos 60 - F_{BH} \times 12\sin 60° - 200 \times 4\sin 60° = 0$

Solving we get,

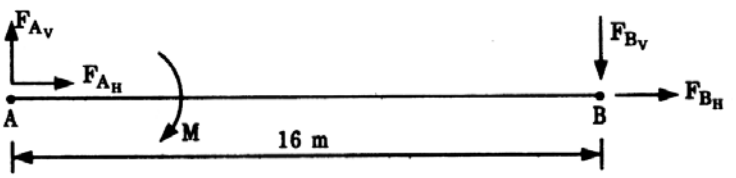
$F_{BV} = -230.9\text{N}$

$F_D = 230.9\text{N}$

Pin Reaction at 'B' $= \sqrt{F_{BV}^2 + F_{BH}^2}$

$= \sqrt{(-230.9)^2 + (-200)^2} = 305.5\text{N}$

FBD of AB :



Taking moments about 'A';

$M + F_{BV} \times 16 = 0$

$\Rightarrow$ M = –(–230.9) × 16 = 3694.4 Nm acting in the clockwise direction

FRICTION

Friction is the opposing force that is set up between the surfaces of contact, when one body slides or rolls or tends to move on the surface of another body.

Coefficient of Friction :
The ratio of Frictional force and Normal Reaction is called coefficient of friction.

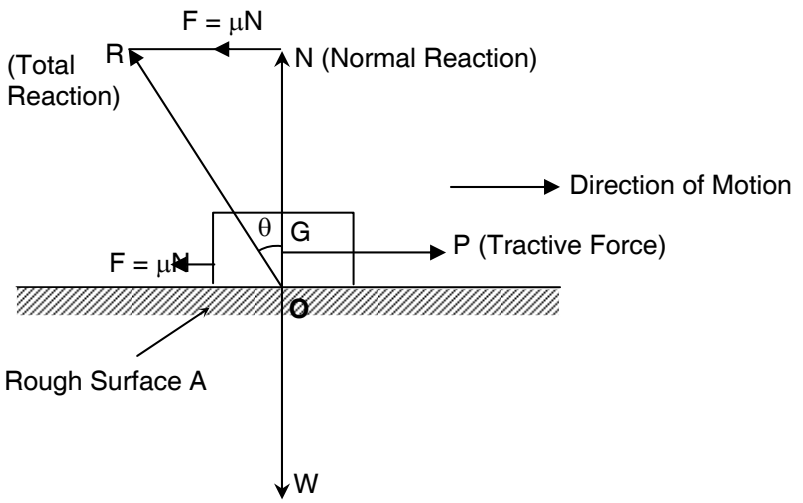
$$\mu = \frac{F}{N}$$

Types of Friction :

i) Sliding Friction :
The friction that exists when one surface slides over another is called sliding friction.

ii) Rolling Friction :
The friction that exists between surfaces which have the balls or the rollers interposed between them is called the rolling friction.

Total Reaction :
Considering the figure below, the block B is in equilibrium under the action of the forces :



i) Its weight W, acting vertically downwards through its e.g. G,
ii) Normal reaction, N of A on B,
iii) Frictional force; $F = \mu N$
where μ is the co–efficient of friction between the surfaces A and B, and
iv) The tractive force P

The force F and N which are at right angles to each other may be combined into a single force

$$R = \sqrt{F^2 + N^2}$$

which is called the total reaction or resultant reaction or sometimes simply the reaction, between the surfaces A and B. If figure, R denotes the rotate reaction.

Angle of Friction :

Let the angle made by the reaction R with the normal reaction N be θ. If F is the frictional force, then at any instant

$$\tan\theta = \frac{F}{R}$$

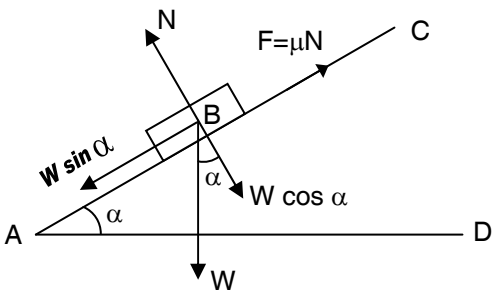
This angle θ increases as F increases. Let ϕ be the maximum value of θ when F has reached the maximum value F_{max}

then $\tan\phi = \frac{F_{max}}{N} = \mu$

ϕ is called angle of friction.

Angle of Repose :

Consider a block B resting on an a plane AC inclined at an angle α to the horizontal AD. The block is in equilibrium under the action of forces as shown :



In the limiting condition when the block is about to slide down the inclined plane AC, the frictional force opposes the motion of block and hence acts up the plane. For equilibrium of the forces along and perpendicular to the plane :

$$F = W\sin\alpha$$

$$N = W\cos\alpha$$

$$\Rightarrow \tan\alpha = \frac{F}{N} = \mu = \tan\phi$$

$$\Rightarrow \alpha = \phi$$

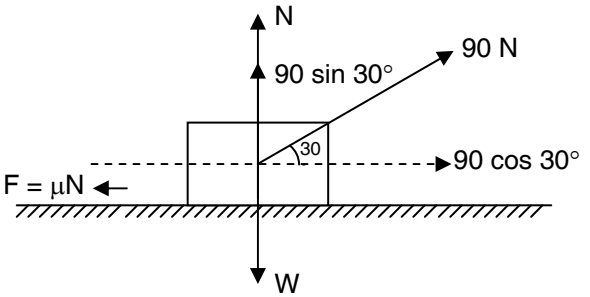
The angle α of the inclined plane AC at which the block B is about to slide down the plane is called the angle of repose, and is equal to the angle of friction ϕ between the block B and the inclined plane AC.

Example 10:

A body resting on a rough horizontal plane requires a pull a force of 90N inclined at 30° to the plane just to move it. It was also found that a push of 110 N inclined at 30° to the plane just moved the body. Determine the weight of the body and the coefficient of friction.

Solution:

Let W = weight of body in Newtons



Now $\Sigma V = 0 \Rightarrow N + 90 \sin 30° - W = 0$ …(1)

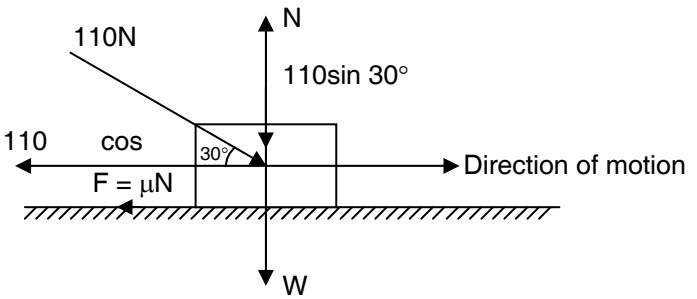
$\Sigma H = 0 \Rightarrow 90\cos 30° - F = 0$

$\Rightarrow 90\cos 30° - \mu N = 0$ …(2)

From (1) and (2) we get,

$$\mu = \frac{90\cos 30°}{W - 90\sin 30°} \quad \text{…(3)}$$

Taking the case when the body requires a push of 110 N at 30° to the plane.



$\Sigma V = 0 \Rightarrow N - W - 110\sin 30° = 0$ …(4)

$\Sigma H = 0 \Rightarrow 110\cos 30° - F = 0$

$\Rightarrow 110\cos 30° - \mu N = 0$ …(5)

From (4) and (5) we get,

$$\mu = \frac{110\cos 30°}{W + 110\sin 30°} \quad \text{…(6)}$$

From (6) and (3) we get; W = 495 N

Substituting W = 495 N in equation (3) we get

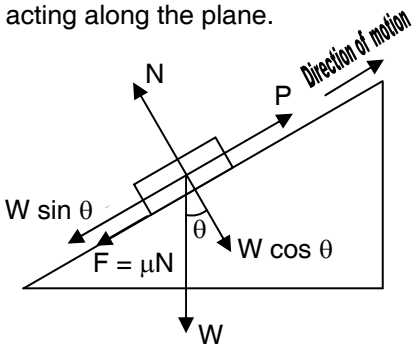
$\mu = 0.1732$

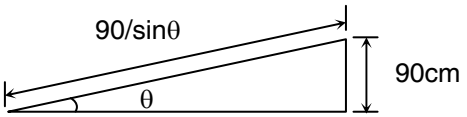
Example 11:

A plane is inclined to horizontal. The work done by the least force in dragging a weight of 50N up the plane through a vertical distance of 90cm is 90 N–m μ = 0.2. Find the angle of inclination of the plane to the horizontal.

Solution :

Let θ be the inclination of the plane to the horizontal and let P be the applied force acting along the plane.





$\Sigma V = 0$ gives $\quad N = W\cos\theta$ …(1)

$\Sigma H = 0$ gives $\quad P = F + W\sin\theta = \mu N + W\sin\theta = \mu W\cos\theta + W\sin\theta$ …(2)

Now $W_d = 90$ N–m $= P \times S = P \times \dfrac{0.9}{\sin\theta}$ $\qquad \left[S = \dfrac{90}{\sin\theta}\text{cm} = \dfrac{0.9}{\sin\theta}\text{m}\right]$

$\Rightarrow \quad 90 = (\mu W\cos\theta + W\sin\theta) \times \dfrac{0.9}{\sin\theta}$

$\Rightarrow \quad 90 = (0.2 \times 50 \times \cos\theta + 50 \times \sin\theta) \times \dfrac{0.9}{\sin\theta}$

$\Rightarrow \quad 100 = \dfrac{10\cos\theta + 50\sin\theta}{\sin\theta}$

$\Rightarrow \quad 10\cot\theta + 50 = 100$

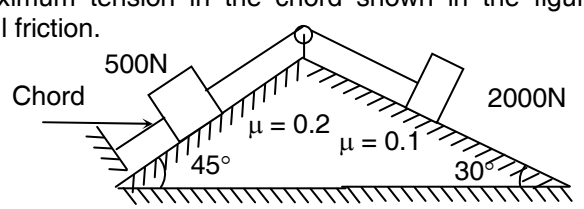
$\Rightarrow \quad 10\cot\theta = 50$

$\Rightarrow \quad \cot\theta = 5$

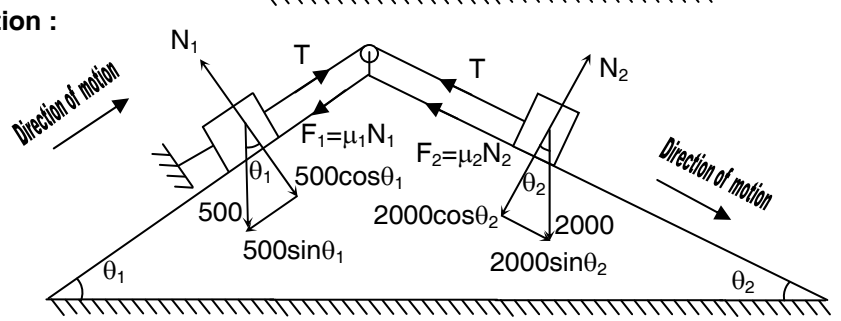
$\Rightarrow \quad \theta = \cot^{-1}(5) = 11°18'$

Example 12 :

Find the maximum tension in the chord shown in the figure, if the bodies have developed full friction.



Solution :



For mass 1;

$\Sigma H = 0$ gives $T = F_1 + 500 \sin \theta_1 = \mu_1 N_1 + 500 \sin \theta_1$

$\Sigma V = 0$ gives $N_1 = 500 \cos \theta_1$

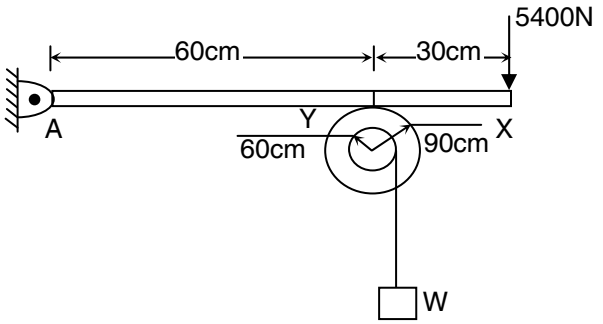
$\Rightarrow$ $T = 500 \sin \theta_1 + \mu_1 \times 500 \cos \theta_1$

$= 500 \sin 45 + 0.2 \times 500 \cos 45 = 424.3$ N

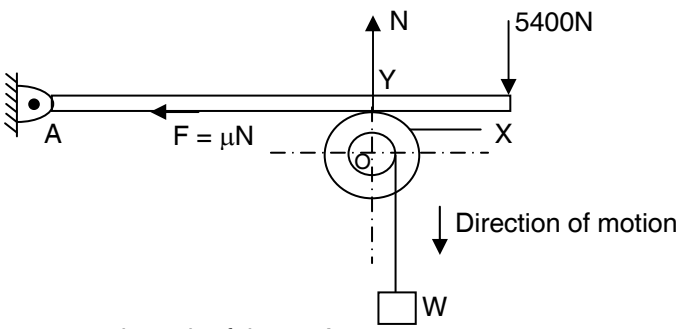
The same result could also be obtained for mass 2.

Example 13 :

What is the maximum load W (Fig) that a 5400 N force will hold up, if the coefficient of friction at Y is 0.25 ? Neglect other friction and weight of the member. If W = 2060 N is suspended from the pulley, what are normal and frictional forces transmitted at Y, taking 4500N force to be applied at X.



Solution :



Taking moments about the fulcrum A;

$N \times 60 = 5400 \times 90$

$\Rightarrow$ $N = 8100$ N

Now $F = \mu N = 0.25 \times 8100 = 2025$N

Taking moments about O;

$W \times 60 = F \times 90$

$\Rightarrow$ $W = \dfrac{2025 \times 90}{60} = 3037.50$N

In the second case when W = 2060N, let F_1 be the frictional force acting at Y. Again taking moments about O:

$F_1 \times 90 = W \times 60$

$\Rightarrow$ $F_1 = \dfrac{2060 \times 60}{90} = 1373.30$N

Also, when the load of 4500N is applied at X, let N_1 be the new value of the normal force at Y. Now taking moments about the fulcrum A,

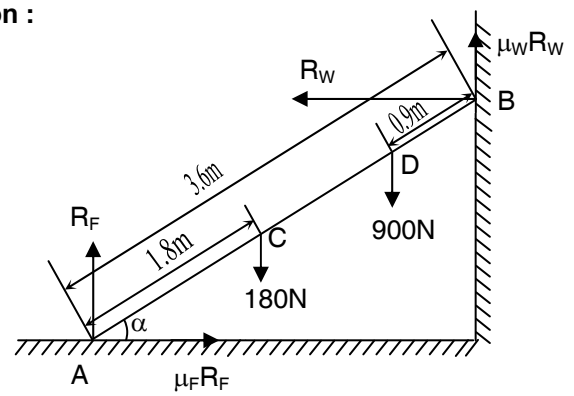
$$N_1 \times 60 = 4500 \times 90$$

$$\Rightarrow \quad N_1 = 6750 \text{ N}$$

Example 14 :

A 3.6 m long ladder 180N in weight is supported against a wall(which is perpendicular to the floor) with its foot on the floor. The coefficient of friction between wall and ladder is 0.2 and between floor and ladder is 0.4. The ladder supports a weight of 900N at a distance of 90 cm from its top. Compute the least value of the angle between the floor and the ladder for its equilibrium.

Solution :



$\Sigma V = 0$ gives $\mu_W R_W + R_F = 180 + 900 = 1080$ …(1)

$\Sigma h = 0$ gives $\mu_F R_F = R_W$

$$\Rightarrow \quad R_F = \frac{R_W}{\mu_F} = \frac{R_W}{0.4} = 2.5R_W \qquad \ldots(2)$$

From (1) and (2)

$$\mu_W R_W + 2.5R_W = 1080$$

$$\Rightarrow \quad R_W = \frac{1080}{2.5 + \mu_W} = \frac{1080}{2.5 + 0.2} = 400\text{N}$$

Taking moments about A we get :

$$180 \times 1.8 \cos\alpha + 900 \times 2.7 \cos\alpha$$

$$= \mu_W R_W \times 3.6 \cos\alpha + R_W \times 3.6 \sin\alpha$$

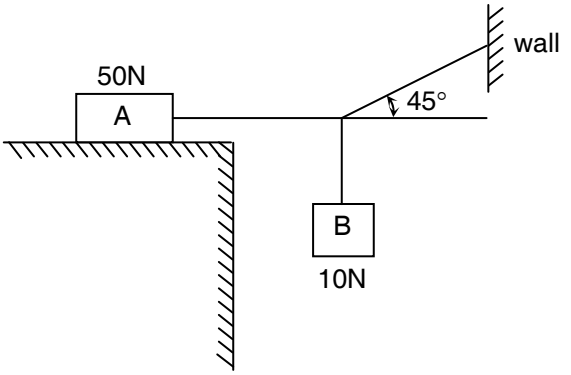
$$\Rightarrow \quad \tan\alpha = \frac{765 - 0.2R_W}{R_W}$$

$$= \frac{765 - 0.2 \times 400}{400} = 1.7125$$

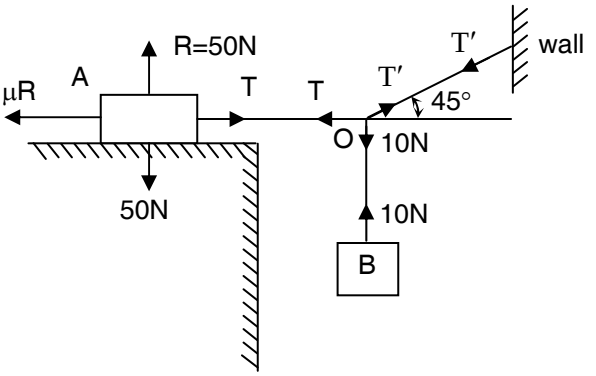
$$\Rightarrow \quad \alpha = \tan^{-1}(1.7125) = 59°43'$$

Example 15 :

In figure shown below the coefficient of static friction between block A and table is 0.2. Find the frictional force exerted on block A. Also find the maximum weight of block B for which the system will be in equilibrium.



Solution :



For the equilibrium of the system we have

$$T = T' \cos 45° \text{ and } T' \sin 45° = 10N$$

From these equations we have T = 10N

Since the block A is in equilibrium

$$R = 50N \text{ and } T = \mu R = 0.2 \times 50 = 10N$$

At this limiting state of equilibrium, all forces all in equilibrium and are maximum.

Thus $W_{max} = 10N$

VIRTUAL WORK

Work :

Work done by a force is given by the dot product or the scalar product of the force and the displacement caused by the force. So, maximum work is done if the displacement produced is along the direction of the force and no work is done if the displacement produced is perpendicular to the direction of force.

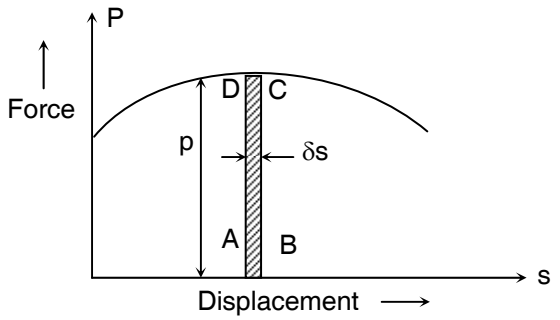
Work done by a Varying Force :

If the magnitude of the force does not remain constant but varies with the displacement as shown in the figure, then work done by the force P during an infinitesimal displacement δs,

$$\delta W = F \cdot \delta s$$
$$= F\,\delta s \cos\theta \quad (\theta \text{ is the angle between F and } \delta s)$$

and hence total work done is,

$$W = \int_0^s \delta W = \int_0^s F \cdot \delta s$$

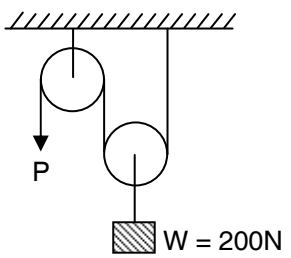


Principle of Virtual Work :

Small possible displacement consistent with the constraints of the system is called virtual displacement. The principle of virtual work or virtual displacement states that the work done by the external forces, on a system of rigid bodies with ideal constraints, during a virtual displacement of the system is equal to zero, if the system is in equilibrium i.e. if δW is the increment of work done by the external forces on the rigid bodies then $\delta W = 0$.

Example 16 :

The figure shows a system of pulleys used to raise a weight of 2000N. Using the method of virtual work, find the force P required to hold the weight in equilibrium.



Solution :

Let the upward moment of weight is y, then virtual work done by W is $-Wy$ (upward movement taken $-$ve)

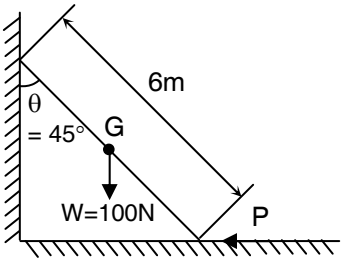
Also, movement of P is 2y downwards, hence virtual work done by P is 2Py.

For equilibrium $\quad 2Py - Wy = 0$

$$\Rightarrow \quad P = \frac{W}{2} = 100\text{N}$$

Example 17 :

A uniform ladder of weight 100N and length 6m is held in equilibrium by a horizontal force P as shown in the figure. Using the method of virtual work, find the value of force P. The ladder makes 45° with the vertical.

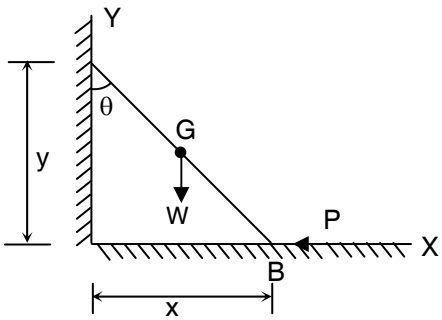


Solution :

Assuming x is measured positive to the right and y positive upwards.

Virtual work done by P for a displacement $\delta x = -P\delta x$.

Virtual work done by W for a corresponding increase $\delta y = -W\delta y$



Applying the principle of virtual work for equilibrium:

$$-P\delta x - W\delta y = 0$$

$$\Rightarrow \quad P\delta x + W\delta y = 0$$

But $x = 6\sin\theta \quad \Rightarrow \quad \delta x = 6\cos\theta\delta\theta$

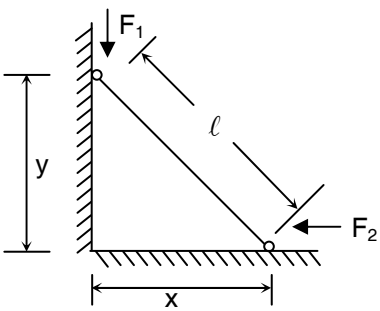
$y = 3\cos\theta \quad \Rightarrow \quad \delta y = -3\sin\theta\delta\theta$

Hence $P(6\cos\theta\delta\theta) + W(-3\sin\theta\delta\theta) = 0$

$$\Rightarrow \quad 2P\cos\theta - W\sin\theta = 0 \quad \Rightarrow \quad P = \frac{W}{2}\tan\theta = \frac{100}{2}\times\tan 45° = 50 \text{ N}$$

Example 18 :

Find the relationship between F_1 and F_2 as shown in the figure, so as to keep this system in equilibrium



Solution :

Length of the bar $\ell = \sqrt{x^2 + y^2}$

$\Rightarrow \quad \ell^2 = x^2 + y^2$

$\Rightarrow \quad 0 = 2x\dfrac{dx}{dy} + 2y \quad \text{or} \quad \dfrac{dy}{dx} = -\dfrac{x}{y}$

Now let the virtual displacement of rod due to F_2 is δx.

Then $\quad \delta y = -\dfrac{x}{y}\delta x$

Work done by $F_2 = -F_2\,\delta x$ (Assuming right and up movement as positive)

Work done by $F_1 = F_1\,\delta y = -F_1\,\dfrac{x}{y}\delta x$

According to the principle of virtual work :

$\delta w = 0$

$\Rightarrow \quad -F_2\,\delta x - F_1\,\dfrac{x}{y}\,\delta x = 0$

$\Rightarrow \quad \dfrac{F_1}{F_2} = -\left(\dfrac{y}{x}\right)$

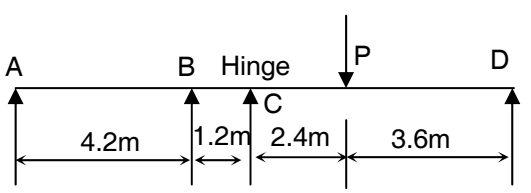
Forces which can be omitted from the equation of virtual work :

The work done of each of the following forces during an infinitesimal virtual displacement, consistent with the constraints imposed on the system, is zero and hence, these forces whenever evolved in some system, may be omitted from the equation of virtual work.

i) Tension of a light inextensible string or the thrust in a rod, whose length is taken as constant.
ii) The internal reaction force between any two bodies of a system which are in contact which each other.
iii) The reaction R of any smooth surface with which the body is in contact.
iv) The reaction of a rough surface on a body which rolls on it without slipping.
v) The reaction at a point or an axis, fixed in space, around which a body is constrained to turn.

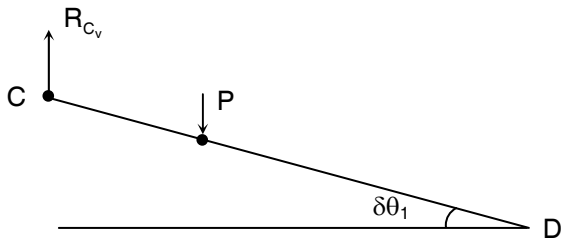
Example 19 :

Two beams AC and CD are hinged at C and are supported at A and D, as shown in the figure. Using the principle of virtual work, determine the reaction at the support B in terms of P.



Solution :

Considering the equilibrium of the beam CD, let it rotate through angle $\delta\theta$, at the hinge C about the support at D as shown in the figure a. As the beam is hinged at C, the reaction at C will have both the vertical as well as the horizontal component. Let R_{C_V} be the vertical component of reaction at C.



Now applying the principle of virtual work,

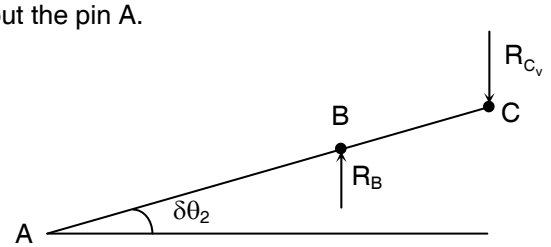
$$P(3.60\delta\theta_1) - R_{C_v} \times \left[(3.60 + 2.40)\delta\theta_1\right] = 0 \qquad \ldots(i)$$

where 3.60 $\delta\theta 1$ and (3.60 + 2.40) $\delta\theta_1$ are respectively the displacements of P and R_{C_v} due to the rotation $\delta\theta_1$.

Thus, from equation (i)

$$R_{C_v} = \frac{3.60}{3.60 + 2.40}P \text{ or } \frac{3}{5}P \qquad \ldots(ii)$$

Now considering the equilibrium of the beam AC, let it rotate through an angle $\delta\theta_2$ about the pin A.



Using the principle of virtual work again,

$$R_{C_v} \times (4.20 + 1.20)\delta\theta_2 - R_B \times (4.20\delta\theta_2) = 0 \qquad \ldots(iii)$$

where (4.20 + 1.20)$\delta\theta_2$ and 4.20 $\delta\theta_2$ are the displacements of R_{C_v} and R_E the reaction at B, respectively.

Thus, from equation (iii)

$$R_B = \frac{(4.20 + 1.20)}{4.20}R_{C_v} = \frac{9}{7}R_{C_v}$$

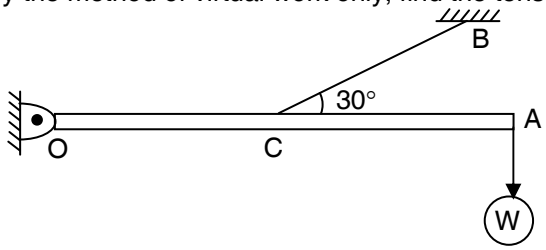
Substituting from equation (ii) $R_{C_v} = \frac{3}{5}P$

$$R_B = \frac{9}{7} \times \frac{3}{5}P = \frac{27}{35}P \text{ or } 0.7714P$$

$\therefore$ Reaction at B = 0.7714P, and acts vertically upwards.

Example 20 :

A beam OA of length ℓ and negligible weight is hinged at O and h as a load W at the free end A. A wire BC supports this beam at the mid point of beam as shown in the figure below. By the method of virtual work only, find the tension in the wire.



Solution :

Let T be the tension in the wire BC. If the weight W is assumed to have a virtual displacement y, then corresponding to this displacement y of the free end, the centre C of the beam will have the displacement $\frac{y}{2}$. The tension T when resolved horizontally and vertically has the components Tcos30° and Tsin30° respectively.

Applying the principle of virtual work :

Total work done = zero

Considering vertically downward direction as positive.

W × (displacement of W) – (T sin 30°) × (displacement at C) = 0

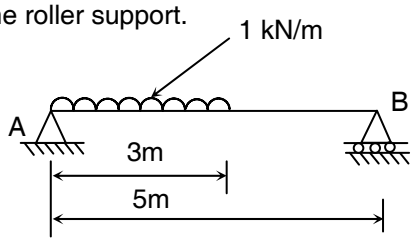
or $\quad W \times y - T\sin 30° \times \frac{y}{2} = 0$

or $\quad \frac{T}{2}\sin 30° = W$

or $\quad T = 4W$

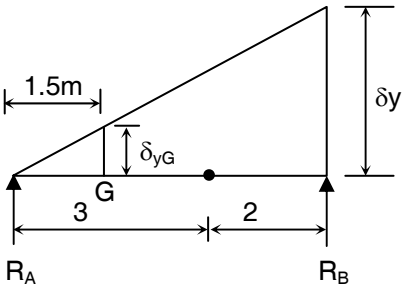
Example 21 :

Figure below shows a simple beam upon which a uniform force of 1KN/m acts. Find the reaction at the roller support.



Solution :

Let the reaction at B be R_B and let δ_y be the vertical displacement at B.



Vertical displacement of C.G.; G of UDL is given by;

$$\delta_{yG} = \delta_y \times \frac{1.5}{5}$$

Now applying the principle of virtual work;

$$\delta Q = 0 \;\Rightarrow\; R_B \times \delta_y - 1 \times 3 \times \left(\delta_y \times \frac{1.5}{5}\right) = 0$$

$$\Rightarrow\; R_B = \frac{1.5 \times 3}{5} = \frac{4.5}{5} = 0.9\text{kN}$$

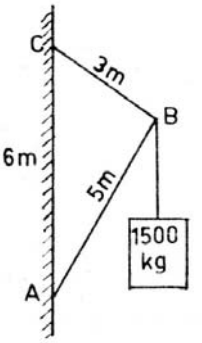
(U.D.L does negative work by moving upwards)

ASSIGNMENT

Duration : 45 Min. **Max. Marks : 30**

Q1 to Q6 carry one mark each

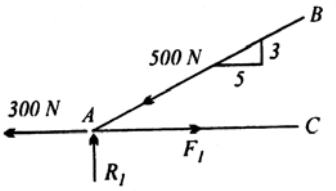
1. Figure shows a hoisting apparatus in which the spar AB is 5m long, is free to turn in a vertical plane through A and B and is fastened by a cable BC 3m long, to a point C, 6 m vertically above. A weight of 1500 kg is supported by a cable at B. The force along AB will be



(A) 1500 kg
(B) 1250 kg
(C) 1000 kg
(D) 750 kg

2. Force along BC will be
(A) 1500 kg (B) 1250 kg
(C) 1000 kg (D) 750 kg

3. Figure shows a roof truss joint and forces acting at the joint



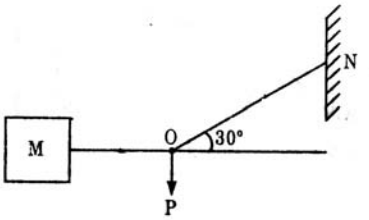
Select the correct statement
(A) F_1 = 557.25 N, R_1 = 428.75 N
(B) F_1 = 728.75 N, R_1 = 428.75 N
(C) F_1 = 728.75 N, R_1 = 257.25 N
(D) F_1 = 257.25 N, R_1 = 557.25 N

4. Which statement is not correct ?
(A) The direction of force from B to A indicates the direction in which the force is acting on the member AB.
(B) The direction of force B to A indicates the direction in which the force is acting upon joint A.
(C) The direction of force from B to A indicates that the member is in compression
(D) The direction of force from B to A indicates that the member AB will exert a force towards the joint at B end of AB.

5. A number of forces acting at a point will be in equilibrium if
(A) their total sum is zero
(B) two resolved parts in two directions at right angles are equal
(C) sum of resolved parts in any two perpendicular directions are both zero
(D) all of them are inclined equally

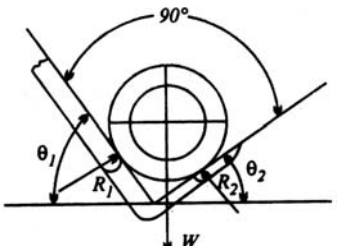
6. A string MON as shown exerts a horizontal force of 100N on the body M. The pull P is

(A) 100 N
(B) 156 N
(C) 200 N
(D) 57.7 N



Q7 to Q18 carry two marks each

7. Pipe of weight W is lifted by a fork lifter whose two mutually perpendicular edges maintain constant angles with horizontal as shown in the figure. Which is the wrong equation among the below?



(A) $R_1 \sin\theta_2 + R_2 \cos\theta_2 = W$
(B) $R_1 \sin\theta_1 + R_2 \cos\theta_2 = 0$
(C) $R_1 \cos\theta_2 - R_2 \cos\theta_1 = 0$
(D) $R_1 \cos\theta_1 + R_2 \sin\theta_1 = W$

8. Which is correct statement ?

(A) $W = R_2 \sin\theta_2$
(B) $W = R_2 \sin\theta_1$
(C) $R_1 = R_2 \dfrac{\sin\theta_2}{\sin\theta_1}$
(D) $R_1 = R_2 \dfrac{\sin\theta_2}{\cos\theta_1}$

9. If $\theta_1 = 30°$, then R_2 is equal to

(A) W
(B) W/2
(C) 2W
(D) 0.866 W

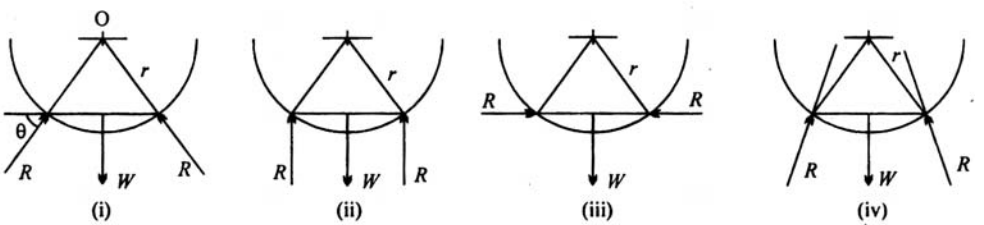
10. If $\theta_2 = 60°$ then the ratio W/R_1 is

(A) 0.5
(B) 0.866
(C) 1.0
(D) 1.1547

11. If $\dfrac{R_1}{R_2} = 1.732$, find the wrong statement

(A) $\theta_1 = \theta_2$
(B) $\theta_1 + \theta_2 = 90°$
(C) $\theta_1 = 30°$
(D) $\theta_2 = 60°$

12. A uniform rod of weight W and length r is supported on the smooth internal surface of a pipe of radius r as shown in the figure



The resultant reaction on the rod is correctly represented by

(A) i (B) ii
(C) iii (D) iv

13. The resultant reaction R makes an angle θ with the axis of rod

(A) $\theta = 30°$ (B) $\theta = 60°$
(C) $\theta = 90°$ (D) $\theta = 45°$

14. The resultant reaction on the pipe is represented by

(A) i (B) ii
(C) iii (D) None

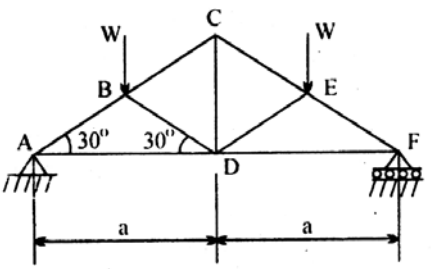
15. The resultant reaction at each end of rod is

(A) W(0.5) (B) W(0.866)
(C) W(0.577) (D) W

16. The reaction on the bar normal to its axis is

(A) 0.5W (B) 0.866 W
(C) 0.577 W (D) W

17. Choose the statement which is not applicable for the roof truss as shown with support and loads



(A) Magnitude of the ratio of the forces in AB and AD is 1.154
(B) Both members AB and AD are subjected to tensile forces
(C) AB is subjected to compression but AD is subjected to tension
(D) Forces in BC and BD are equal in magnitude

18. Force in AD and AB are respectively

(A) 1.733 W and 2W (B) 2 W and 1.733 W
(C) 1.733 W and W (D) 1.733 W and –2W

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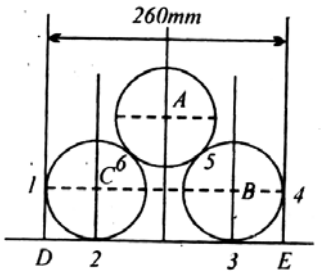
TEST PAPER

Duration : 30 Min. **Max. Marks : 25**

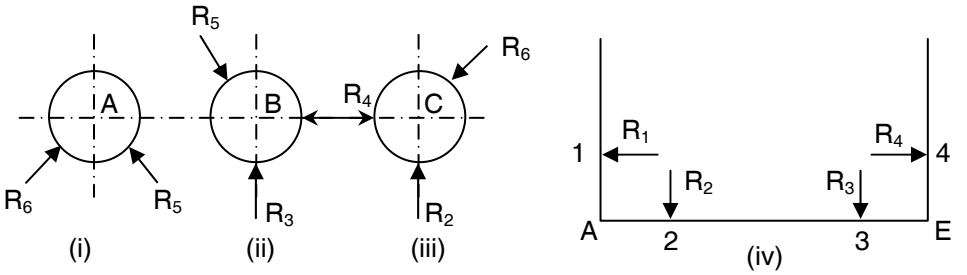
Q1 to Q5 carry one mark each

1. If a number of forces are acting simultaneously on a particle, then each one of them will produce the same effect, which it would have done while acting alone. This is known as
(A) principle of transmissibility of forces
(B) principle of physical independence of forces
(C) principle of resolution of forces
(D) principle of physical dependence of forces

2. Three identical spheres, each weighing 10 N and having a diameter of 100 mm fit inside a smooth box as shown in the figure



The free body diagrams of spheres A, B and C and the box are shown at (i), (ii), (iii) and (iv)



(A) i and ii are correct (B) ii and iii are correct
(C) iii and iv are correct (D) iv and i are correct

3. Which is not correct ?
(A) $R_5 = R_6$ (B) $R_1 = R_4$
(C) $R_2 = R_3$ (D) $R_2 + R_3 = 2W = 20$ N

4. Find the correct statement
(A) AC and AB make right angle at A
(B) A, C and D and A, B and E are collinear
(C) The line of weight of A makes an angle of 53.13° with AC
(D) The reaction R_5 and R_6 make angles of 53.13° each with BC.

5. Which statement is correct ?

(A) $\frac{W}{\sin 106.2} = \frac{R_5}{\sin 126.87}$ (B) $\frac{W}{\sin 53.63} = \frac{R_5}{\sin 73.8}$

(C) $\frac{W}{\sin 126.87} = \frac{R_1}{\sin 90}$ (D) $\frac{W - R_2}{\sin 36.87} = \frac{R_1}{\sin 90}$

Q6 to Q13 carry two marks each

6. Two vectors $\vec{A}$ and $\vec{B}$ are such that

$$\left|\vec{A} + \vec{B}\right| = \left|\vec{A} - \vec{B}\right|$$

Then, the angle between two vectors $\vec{A}$ and $\vec{B}$ will be

(A) 0° (B) 60°

(C) 90° (D) 180°

7. The resultant of two forces P and Q is R. If one of the forces is reversed in direction, the resultant becomes S. Then, for the identity $R^2 + S^2 = 2\left(P^2 + Q^2\right)$ to hold good

(A) force can have any angle of inclination between them

(B) forces act at right angles to each other

(C) forces are inclined at 45° to each other

(D) forces are collinear

8. Given that $\vec{A} + \vec{B} = \vec{A} - \vec{B}$. This can be true, when

(A) $\vec{A} = \vec{0}$ (B) $\vec{B} = \vec{0}$

(C) neither $\vec{A}$ nor $\vec{B}$ is a null vector (D) $\vec{A}$ is perpendicular to $\vec{B}$

9. Which of the following sets of concurrent forces may be in equilibrium?

(A) $F_1 = 3\text{kgf}, F_2 = 5\text{kgf}, F_3 = 9\text{kgf}$

(B) $F_1 = 3\text{kgf}, F_2 = 5\text{kgf}, F_3 = 1\text{kgf}$

(C) $F_1 = 3\text{kgf}, F_2 = 5\text{kgf}, F_3 = 15\text{kgf}$

(D) $F_1 = 3\text{kgf}, F_2 = 5\text{kgf}, F_3 = 6\text{kgf}$

10. Let the angle between two non–zero vectors, $\vec{A}$ and $\vec{B}$ be 120° and its resultant be $\vec{C}$. Then,

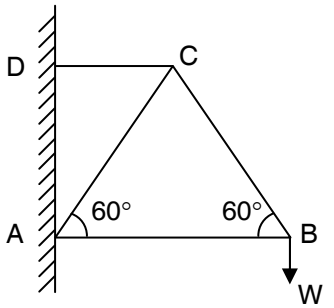
(A) C must be equal to $|A - B|$

(B) C must be less than $|A - B|$

(C) C must be greater than $|A - B|$

(D) C may be equal to $|A - B|$

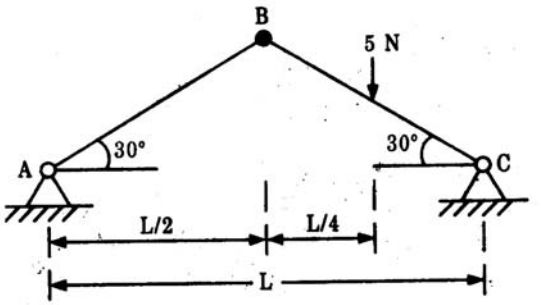
11. In a framed structure, as shown in figure, the force in the member BC is



(A) $\frac{W}{\sqrt{3}}$ (compression) (B) $\frac{W}{\sqrt{3}}$ (tension)

(C) $\frac{2W}{\sqrt{3}}$ (compression) (D) $\frac{2W}{\sqrt{3}}$ (tension)

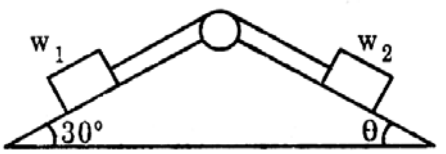
12. ABC is a three pinned arch. If the direction of the resultant reaction at the support A is at θ degrees to the horizontal then ;



(A) $\theta = 60°$ (B) $\theta = 45°$

(C) $\theta = 90°$ (D) $\theta = 30°$

13. Given W_1 = 400 N and W_2 = 300 N. Planes are smooth for the system shown. For equilibrium, θ and tension in the string are



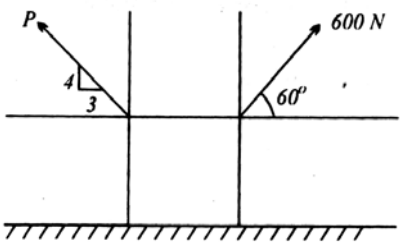
(A) 41.8°, 200 N (B) 20°, 100 N

(C) 45°, 50 N (D) 60°, 60 N

Q14(a) & (b) carry two marks each

Linked Answer Question

14(a). A block is subjected to two forces as shown in the figure



If the horizontal resultant force is required to be zero, P will be

(A) 375 N (B) 500 N

(C) 649.5 N (D) 866 N

14(b). In the above part, if the resultant force along horizontal is zero the vertical resultant force is

(A) 919.6N (B) 866 N

(C) 649.5 N (D) 620 N

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